Precarious necklaces

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Abstract

A necklace of coloured beads is precarious if it has no rotational symmetry, yet removing all the beads of any one colour leaves a necklace that does. We show that the shortest precarious necklace with k colours has length $2m\,P_m$ when $k = 2m$ and $P_m(2m\,p_{m+1} + 1)$ when $k = 2m + 1$, where p_i is the i-th prime and $P_m = p_1 \cdots p_m$. The same method answers the whole family of questions in which one keeps any s of the k colours: the minimum is $s\,P_{\lfloor s/2 \rfloor + 1} + (k - s)\,P_{\lceil s/2 \rceil}$. The key fact is that a word is a q-th power as soon as all its two-letter projections are. Every result is formalised in Lean.

1 The question

A necklace is a word written around a circle, so that rotating it does not change it. The necklace `AABB` has no rotational symmetry: no rotation short of a full turn brings it back to itself. Remove the two `A`s and what is left, `BB`, does have a symmetry, and the same happens if the `B`s are removed. Call a necklace with at least two colours *precarious* if it has no rotational symmetry but removing all the beads of any single colour leaves a necklace that has one. Philip Weiss asked on MathOverflow [3] for the length $a(k)$ of the shortest precarious necklace with k colours, and gave the first values

$$a(2) = 4, \qquad a(3) = 14, \qquad a(4) = 24, \qquad a(5) = 126,$$

with `ABCCBBCACBBCCB` as a shortest example for $k = 3$. Before reading on, it is worth trying to find a precarious necklace with three colours. Short attempts fail in an instructive way: making one deletion symmetric tends to make another one asymmetric.

The four values do not suggest a pattern, and the jump from 24 to 126 is the first hint that primes are involved. Here is the answer. Write $p_1 = 2$, $p_2 = 3$, $p_3 = 5, \ldots$ for the primes and $P_m = p_1 p_2 \cdots p_m$ for the product of the first m of them, with $P_0 = 1$.

Theorem 1. *For every $m \geq 1$,*

$$a(2m) = 2m\,P_m, \qquad a(2m + 1) = P_m\,(2m\,p_{m+1} + 1).$$

The sequence continues $180, 1290, 1680, 18690, 23100, \ldots$, so it grows faster than any exponential.

The proof rests on one observation about words, Lemma 3 below. Whether a word is a q-th power is decided by its two-letter projections. With that in hand the problem becomes a question about how primes can be shared out among the colours, and the same argument settles a whole family of variants. Removing one colour from k is the same as keeping $k - 1$ of them. More generally, call a necklace *s-precarious* if it has no rotational symmetry but keeping the beads of any s of its k colours, and removing the rest, leaves a necklace with one.

Theorem 2. *Let $1 \leq s < k$. The shortest s-precarious necklace with k colours has length*

$$s\,P_{\lfloor s/2 \rfloor + 1} + (k - s)\,P_{\lceil s/2 \rceil}. \tag{1}$$

Read (1) as two kinds of colour. In a shortest example, s of the colours are used $P_{\lfloor s/2 \rfloor + 1}$ times each and the other $k - s$ are used $P_{\lceil s/2 \rceil}$ times each. When s is odd the two products agree and every colour is used equally often. For $s = k - 1$ the formula is Theorem 1. At the other end, $s = 1$ gives $2k$ (every colour must appear at least twice), $s = 2$ gives $2k + 8$ and $s = 3$ gives $6k$. Table 1 lists small cases.

$k\backslash s$	1	2	3	4	5	6
3	6	14				
4	8	16	24			
5	10	18	30	126		
6	12	20	36	132	180	
7	14	22	42	138	210	1290

Table 1: The minimum length (1). The diagonal $s = k - 1$ is the original question.

2 Words, rotations and projections

We work with words, read linearly, and their letters (the colours). A necklace is a word up to rotation. For a word w and a set S of letters, the projection w_S keeps the letters in S and deletes the rest, and w_{xy} is short for $w_{\{x,y\}}$. A word is a *proper power* if it equals $v^r = vv\cdots v$ for some word v and some $r \geq 2$, and a *q-th power* if it equals v^q. We write $|w|$ for the length of w and n_x for the number of occurrences of the letter x.

Lemma 1. *A nonempty word w is fixed by a rotation through s places, for some $0 < s < |w|$, if and only if w is a proper power. Moreover, rotating w rotates each projection w_S.*

Proof. If $w = v^r$ with $r \geq 2$, rotating through $|v|$ places fixes w. For the converse, suppose rotation through s places fixes w and let $d = \gcd(s, |w|)$. Some multiple of s is congruent to d modulo $|w|$, and rotating through a multiple of s also fixes w, so rotation through d places does. Then w has period d, and since d divides $|w|$ and $d \leq s < |w|$, w is the $(|w|/d)$-th power of its first d letters. The last statement holds because deleting letters from a rotation of w gives a rotation of the deleted word. □

So a necklace has a rotational symmetry exactly when the words that represent it are proper powers, and we may forget circles and work with powers of words. One consequence is used repeatedly.

Lemma 2. *If $w_S = v^r$, then r divides n_x for every $x \in S$.*

Proof. Each copy of v contains the same number of xs. □

The next lemma is the heart of the matter. We could not find it in the literature, although it is simple enough that it may well be known.

Lemma 3. *Let w be a word with at least two letters and let $q \geq 1$. If w_{xy} is a q-th power for every pair of distinct letters x, y, then w is a q-th power.*

The proof uses a picture. Reading w from the left and counting the letters seen so far traces a path in $\mathbb{Z}^k$ from the origin to $(n_x)_x$, one unit step at a time. Projecting the path onto the plane of two coordinates x, y gives the path of w_{xy}, with pauses at the steps that use other letters. Figure 1 shows one such shadow. If $w_{xy} = v^q$, its path is q copies of the same staircase, each shifted by $(n_x/q, n_y/q)$ from the last.

Proof. Each $w_{xy} = v^q$ shows that q divides n_x; put $s_x = n_x/q$ and $B = \sum_x s_x = |w|/q$. Work with $W = ww$, whose two-letter projections are $2q$-th powers of the same words, and let $f(t) \in \mathbb{Z}^k$ count the letters among the first t letters of W. The points $f(0), f(1), \ldots$ form a chain: if $t < t'$ then $f(t) \leq f(t')$ in every coordinate. Fix $t \leq |w|$ and put $p = f(t) + s$. We show $f(t + B) = p$.

First, every shadow of p lies on the corresponding shadow path. The point $(f_x(t), f_y(t))$ is the count of xs and ys in a prefix of W_{xy} of length $j \leq |w_{xy}|$, and in a $2q$-th power of v the prefix

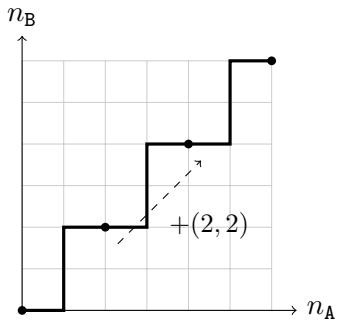


Figure 1: The shadow of the necklace of Figure 2 on the $\mathtt{A},\mathtt{B}$ plane. The projection is $w_{\mathtt{AB}} = (\mathtt{ABBA})^3$, so the path repeats after every shift by $(2,2)$.

of length $j + |v|$ contains one more copy's worth of letters, so it has counts (p_x, p_y). That prefix of W_{xy} comes from a prefix of W, so $(p_x, p_y) = (f_x(t'), f_y(t'))$ for some t'.

Now suppose $f(t + B) \neq p$. Both have coordinate sum $t + B$, so some coordinate x has $f_x(t+B) > p_x$ and another coordinate y has $f_y(t+B) < p_y$. Take t' with $(f_x(t'), f_y(t')) = (p_x, p_y)$. Since $f_x(t') < f_x(t + B)$, the chain property forces $t' < t + B$, and then $f_y(t') \leq f_y(t + B) < p_y$, a contradiction.

So $f(t + B) = f(t) + s$ for every $t \leq |w|$. Comparing the steps at t and $t + B$ shows that the letter in position $t + B$ of w equals the letter in position t whenever both exist. Hence w has period B, and as $|w| = qB$, it is the q-th power of its first B letters. □

3 Constructions

The upper bounds come from interleaving pairs of colours, each pair with a rhythm of its own. Figure 2 shows the case $k = 4$, using Weiss's example

$$\mathtt{ABCD\ BACD\ ABDC\ BACD\ ABCD\ BADC}.$$

It has six blocks. In each block the pair $\mathtt{A},\mathtt{B}$ appears as $\mathtt{AB}$ or $\mathtt{BA}$, alternating, so its pattern has period 2; the pair $\mathtt{C},\mathtt{D}$ appears as $\mathtt{DC}$ in every third block and as $\mathtt{CD}$ otherwise, period 3. Removing $\mathtt{C}$ or $\mathtt{D}$ destroys the second rhythm and leaves a word of period 2 blocks, a cube. Removing $\mathtt{A}$ or $\mathtt{B}$ leaves period 3 blocks, a square. The necklace itself only repeats when both rhythms line up, after $\mathrm{lcm}(2,3) = 6$ blocks, which is the whole necklace.

block 0	block 1	block 2	block 3	block 4	block 5
A B C D	B A C D	A B D C	B A C D	A B C D	B A D C

Figure 2: The shortest precarious necklace with four colours, read left to right and closed into a circle. The shaded pair $\mathtt{A},\mathtt{B}$ swaps every block; the pair $\mathtt{C},\mathtt{D}$ is reversed only in blocks 2 and 5.

In general give the i-th pair of colours, $\{x_i, y_i\}$, the prime p_i, and let block j list the pairs in order, pair i as $x_i y_i$ if p_i divides j and as $y_i x_i$ otherwise. (Weiss's example follows the mirror rule for the second pair, which works equally well.)

Odd $s = 2h + 1$. Use $h + 1$ pairs and P_{h+1} blocks, and put the remaining $k - 2h - 2$ colours at the end of every block, in a fixed order. Each colour appears once per block, so the length is $k\,P_{h+1}$. A set of s colours cannot contain all $2h + 2$ pair colours, so it misses a colour of some pair i. Keeping only that set, what remains of pair i no longer depends on the block, and the blocks repeat with period P_{h+1}/p_i: the kept word is a p_i-th power. The whole word is not a proper power. A rotation fixing it moves every colour by the same number of blocks, since each

colour occurs once per block, and pair i only returns to its pattern after a multiple of p_i blocks, so the shift is a multiple of P_{h+1}.

Even $s = 2h$. Use h pairs and $P_h\,p_{h+1}$ blocks, so the indices j run up to $P_h\,p_{h+1} - 1$, and put a cluster of the remaining $k - s$ colours, in a fixed order, before each group of p_{h+1} consecutive blocks. There are P_h groups, each of length $k - s + 2h\,p_{h+1}$, so the length is $P_h(k-s) + 2h\,P_{h+1} = s\,P_{h+1} + (k-s)\,P_h$. A set of s colours either misses a colour of some pair i, and then the groups repeat with period P_h/p_i, or it consists of exactly the $2h$ pair colours, and then it keeps the blocks alone, which repeat with period P_h blocks. For the whole word, suppose it equals v^r with $r \geq 2$. A cluster colour occurs P_h times, so some prime factor p of r divides P_h, say $p = p_i$. Pair i would then repeat after $P_h\,p_{h+1}/p_i$ blocks, which is impossible because p_i divides neither P_h/p_i nor p_{h+1}.

For $s = k - 1$ these are the constructions behind Theorem 1: pairs alone when k is even, and a single extra colour before every p_{m+1} blocks when $k = 2m + 1$. Weiss's example for $k = 5$ has exactly this shape, with six groups of five blocks.

Remark. For $s = k - 1$ the constructions satisfy more than is asked: removing any nonempty set of colours, not just one, leaves a symmetric necklace. So requiring every deletion to be symmetric does not change $a(k)$.

4 The lower bound

Let N be an s-precarious necklace with k colours, $1 \leq s < k$, and let $\mathcal{K}$ be the set of s-element sets of colours. For each $S \in \mathcal{K}$ the projection N_S is a proper power v^r; choose a prime $c(S)$ dividing its exponent. By Lemma 2, $c(S)$ divides n_x for every $x \in S$.

Avoided pairs. Fix a prime q. If every pair of colours lay in some S with $c(S) = q$, then by Lemma 2 each two-letter projection would be a projection of a q-th power, hence a q-th power itself, and by Lemma 3 N would be a q-th power, contradicting Lemma 1. So each prime q has an *avoided pair* e_q of colours that lies in no S with $c(S) = q$.

Spanning. For a colour x, let Q_x be the set of primes $c(S)$ over the sets $S \ni x$. Then n_x is divisible by every prime in Q_x, so $n_x \geq P_{|Q_x|}$, since distinct primes multiply to at least the product of the same number of smallest primes. Let U_x be the union of $\{x\}$ and the avoided pairs e_q, $q \in Q_x$. If $|U_x| \leq s$, some $S \in \mathcal{K}$ contains U_x; then $x \in S$, so $c(S) \in Q_x$, and S contains the avoided pair of $c(S)$, which is impossible. Therefore

$$s + 1 \leq |U_x| \leq 1 + 2|Q_x|. \tag{2}$$

Odd $s = 2h + 1$. By (2), $|Q_x| \geq h + 1$ for every colour, so $|N| = \sum_x n_x \geq k\,P_{h+1}$, which is (1).

Even $s = 2h$. Now (2) only gives $|Q_x| \geq h$. Call x *heavy* if $|Q_x| \geq h + 1$ and *light* otherwise. A colour y that lies in some S whose prime's avoided pair contains y is heavy: then y already belongs to one of its own avoided pairs, and the first inequality in (2) improves to $s + 1 \leq 2|Q_y|$.

Suppose x is light. Then both inequalities in (2) are equalities: the h avoided pairs of Q_x are disjoint, they miss x, and together with x they form a set V of $s + 1$ colours. Take $y \in V \setminus \{x\}$, lying in the avoided pair e_q, and let z be its partner in that pair. The set $V \setminus \{z\}$ belongs to $\mathcal{K}$ and contains x, so its prime lies in Q_x, and its prime's avoided pair is not inside it. Every avoided pair of Q_x other than e_q is inside it, so its prime is q. Since $y \in V \setminus \{z\}$ and $y \in e_q$, the colour y is heavy.

Thus a single light colour forces the s colours of $V \setminus \{x\}$ to be heavy. If there is a light colour, $|N| \geq s\,P_{h+1} + (k-s)\,P_h$; if there is none, $|N| \geq k\,P_{h+1}$, which is larger. This is (1), and together with the constructions it proves Theorem 2. Theorem 1 is the case $s = k - 1$: for $k = 2m$ the formula reads $(2m-1)P_m + P_m = 2m\,P_m$, and for $k = 2m + 1$ it reads $2m\,P_{m+1} + P_m$.

5 Formal verification

Every statement above has been checked in the Lean 4 proof assistant [1] with the Mathlib library [2], using only Lean's standard axioms. The formalisation covers Lemmas 1 to 3, both constructions for all k and s, the lower bound, Theorems 1 and 2, the identification of precarious with $(k-1)$-precarious, and the remark. As a check that the statements say what they should, each of 39 small changes to them (an off-by-one in a bound, a wrong exponent, a swapped hypothesis) was confirmed to make the file fail to compile. The Lean files, the mutation tests and the programs that check the constructions and Table 1 independently are at [4].

Acknowledgements

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